DO NOW — RECALL

# A*: heuristic reliability and route evaluation

Name: _____ Class: _____ Date: _____

**1. Calculate f when g=4 and h=3.**

**2. What is the role of a predecessor table?**

**3. If S–A=2 and A–G=5, what is this route's total cost?**

**4. The true cheapest remaining cost from A is 4. Is h(A)=6 admissible?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

f=7.

### 2. Model answer

It lets us reconstruct the path by following predecessors backwards.

### 3. Model answer

7.

### 4. Model answer

No. It overestimates the true cost.

### Vocabulary — know these terms

admissible — Heuristic never overestimates the true cheapest remaining cost.

consistent — Heuristic obeys h(u) ≤ edge(u,v)+h(v) for every edge.

suboptimal — A valid route whose cost is greater than the cheapest possible.

frontier — Discovered vertices awaiting expansion.

heuristic — An estimated remaining cost used to guide search.

NEW LEARNING

# Read, annotate and explain

### Find the real remaining cost

From A, the cheapest route A→B→G costs 2+2=4. Therefore h(A)=3 is admissible, but h(A)=9 is not.

### Safe estimates support optimal search

With h(A)=3, A is expanded before B, improving the route to B and finally G at cost 6.

### Watch the misleading estimate

With h(A)=9, A has f=11; B has f=7 and leads to goal at cost 8.

### Why the error matters

A* that stops on the first goal removed from the open set can return this 8-cost path before exploring A, even though a 6-cost route exists.

### Check the guarantee

Admissibility means no overestimate. Consistency adds the edge test h(u) ≤ weight(u,v) + h(v). Without reopening settled nodes, A* requires consistency for its shortest-path guarantee. Explain this as well as the route cost.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

### Check admissibility and trace U→G with safe estimates.

### Predict the choice

V has g1,h3,f4; W has g5,h1,f6. Expand V.

### Review the choice

V improves W to g3 and f4; expand W.

### Review the choice

W improves the goal G to g5,h0,f5.

### Review the choice

The path is U→V→W→G with total cost 5; the heuristics do not exceed the actual remaining costs.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

The true remaining distance from node A to goal G is 6. Decide whether h(A)=4 and h(A)=8 are admissible, with reasons.

### Question 2

Edges S–A=2, S–B=6, A–B=2, B–G=2, A–G=8. Calculate the best S→G route and explain why h(A)=9 can mislead A*.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare Dijkstra's algorithm and A* in their selection rule, heuristic use and guarantee of shortest paths.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**If a heuristic overestimates the true remaining cost, A* may lose its shortest-route guarantee.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.